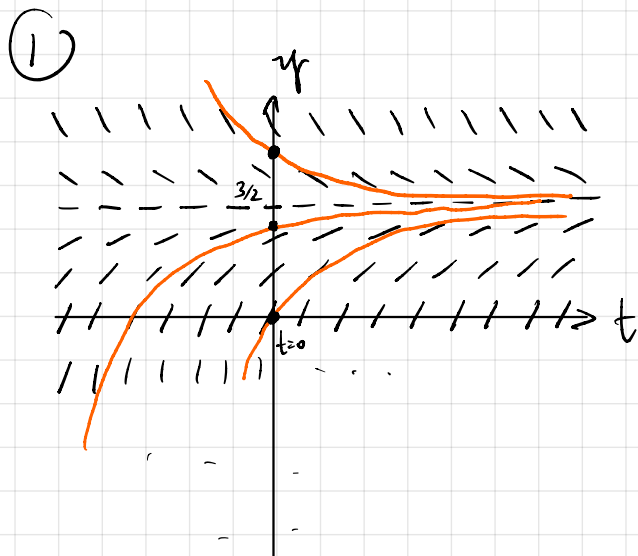


202 PS 1

In each of Problems 1 through 6, draw a direction field for the given differential equation. Based on the direction field, determine the behavior of y as $t \rightarrow \infty$. If this behavior depends on the initial value of y at $t = 0$, describe the dependency.

1. $y' = 3 - 2y$

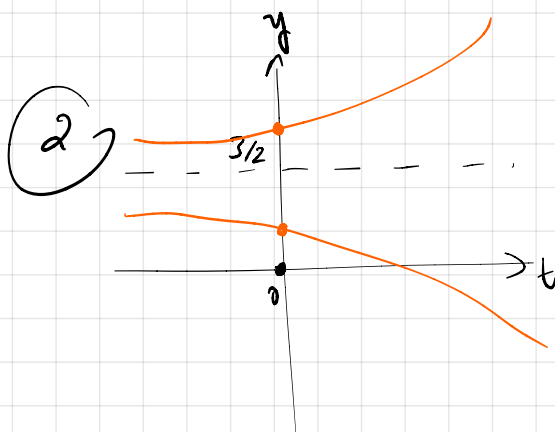
2. $y' = 2y - 3$



$y_0 = y(0) > 3/2 \quad \text{as } t \rightarrow \infty \quad y \rightarrow 3/2$

$y(0) < 3/2 \quad \text{as } t \rightarrow \infty \quad y \rightarrow 3/2$

$y(0) = \frac{3}{2} \Rightarrow y = \frac{3}{2} \quad \text{as } t \rightarrow \infty \quad y = \frac{3}{2}$



$y(0) > \frac{3}{2} \quad y \rightarrow \infty \quad \text{as } t \rightarrow \infty$

$y(0) < \frac{3}{2} \quad y \rightarrow -\infty \quad \text{as } t \rightarrow \infty$

$y(0) = \frac{3}{2} \Rightarrow y \rightarrow \frac{3}{2} \quad \text{as } t \rightarrow \infty$

In each of Problems 1 through 6, determine the order of the given differential equation; also state whether the equation is linear or nonlinear.

1. $t^2 \frac{d^2 y}{dt^2} + t \frac{dy}{dt} + 2y = \sin t$

2. $(1 + y^2) \frac{d^2 y}{dt^2} + t \frac{dy}{dt} + y = e^t$

3. $\frac{d^4 y}{dt^4} + \frac{d^3 y}{dt^3} + \frac{d^2 y}{dt^2} + \frac{dy}{dt} + y = 1$

4. $\frac{dy}{dt} + ty^2 = 0$

5. $\frac{d^2 y}{dt^2} + \sin(t + y) = \sin t$

6. $\frac{d^3 y}{dt^3} + t \frac{dy}{dt} + (\cos^2 t)y = t^3$

PDE/
ODE

lin/non
lin.

order

- | | | | |
|----|-----------------|-------|---------|
| 1) | 2 nd | order | lin. |
| 2) | 2 nd | order | nonlin. |
| 3) | 4 th | order | lin. |
| 4) | 1 st | order | nonlin. |
| 5) | 2 nd | order | nonlin. |
| 6) | 3 rd | order | lin. |

ODE

$\mathcal{L}[y] - \text{lin?}$

In each of Problems 21 through 24, determine the order of the given partial differential equation; also state whether the equation is linear or nonlinear. Partial derivatives are denoted by subscripts.

21. $u_{xx} + u_{yy} + u_{zz} = 0$

23. $u_{xxxx} + 2u_{xxyy} + u_{yyyy} = 0$

22. $u_{xx} + u_{yy} + \underline{uu_x} + \underline{uu_y} + u = 0$

24. $u_t + \underline{uu_x} = 1 + u_{xx}$

) PDE

21) 2nd ord. lin
 22) 2nd ord. nonlin
 23) 4th ord. lin
 24) 2nd ord. nonlin.

$$xy u_{xx} + 4u_y + tz u_{yy} = 0$$

dep. var. $\nearrow$
 $\nwarrow$ indep. var.s
 $u(x, y, z, t)$

$= \sin u$ nonlin.

$u \cdot u_{xy}$ nonlin
 u^2 nonlin.

e^t lin
 e^u nonlin.

(c) Find the general solution of the given differential equation, and use it to determine how solutions behave as $t \rightarrow \infty$.

5. $y' - 2y = 3e^t$

10. $y' + 2y = te^{-2t}$, $y(1) = 0$

1st ord. lin $\Rightarrow \mu$ integr. factor

5) $y' - 2y = 3e^t$ • $\mu(t) \Rightarrow \int \frac{d}{dt}(\mu y) dt = \int 3e^t \cdot \mu(t) dt$ $\mu(t)$

$\cdot \mu(t) \left(\begin{aligned} \mu \cdot y' - 2\mu y &= 3e^t \mu \\ \underline{\mu \cdot y'} - 2\mu y &= 3e^t \mu \\ &= (\mu \cdot y)' \text{ if } \underline{\mu'} = 2\mu \end{aligned} \right.$

$$\mu y = I(t)$$

$$y = \frac{I(t)}{\mu(t)}$$

$$\mu' = -2\mu$$

$$\frac{\mu'}{\mu} = -2$$

$$\int \frac{\mu'}{\mu} dt = -\int 2 dt$$

$$\ln|\mu| = -2t + C$$

$$e^{\ln|\mu|} = e^{-2t+C}$$

$$|\mu| = e^{-2t} \cdot \underbrace{e^C}_{\in \mathbb{R} > 0} \Rightarrow \mu(t) = C e^{-2t}$$

$$\underbrace{C e^{-2t} y' - 2 C e^{-2t} y}_{(e^{-2t} y)'} = 3 e^t \cdot C e^{-2t}$$

$$\int \frac{d}{dt} (e^{-2t} y) dt = \int 3 e^{-t} dt$$

$$e^{-2t} y = -3 e^{-t} + C \quad \cdot e^{2t}$$

$$y = -3 e^t + C \cdot e^{2t} = \underbrace{e^{2t}}_{\rightarrow \infty} (-\underbrace{3 e^{-t}}_{\rightarrow 0} + C)$$

$$(10) \quad y' + 2y = t e^{-2t}, \quad y(1) = 0$$

$$C > 0 \Rightarrow y \rightarrow \infty$$

$$C \leq 0 \Rightarrow y \rightarrow -\infty$$

$$\mu y' + \underbrace{\mu'}_{2\mu} y = \mu t e^{-2t}$$

$$\mu' = 2\mu \Rightarrow \int \frac{\mu'}{\mu} = \int 2 \Rightarrow \ln|\mu| = 2t + C \Rightarrow |\mu| = e^{2t} \cdot C_{>0}$$

$$\mu = C \cdot e^{2t}$$

$$(\mu y)' = \mu y' + \mu' y = e^{2t} y' + 2e^{2t} y = e^{2t} t e^{-2t} = t$$

$$(e^{2t} y)'$$

$$\int (e^{2t} y)' dt = \int t dt$$

$$e^{2t} y = \frac{t^2}{2} + C \quad (\cdot e^{-2t})$$

$$y = \frac{t^2 e^{-2t}}{2} + C e^{-2t} \rightarrow \text{general soln.}$$

$$\text{IV: } y(1) = \frac{1^2 \cdot e^{-2}}{2} + C \cdot e^{-2} = 0$$

$$C e^{-2} = -\frac{e^{-2}}{2}$$

$$C = -\frac{1}{2}$$

$$\text{Soln to the IVP: } y = \frac{t^2 e^{-2t}}{2} - \frac{e^{-2t}}{2} \rightarrow 0 \text{ as } t \rightarrow \infty$$

$$16. y' + (2/t)y = (\cos t)/t^2, \quad y(\pi) = 0, \quad t > 0$$

1st ord. lin.

$$\mu y' + \underbrace{\mu \frac{2}{t}}_{=\mu'} y = \mu \frac{\cos t}{t^2}$$

$$\mu' = \mu \frac{2}{t} \Rightarrow \int \frac{\mu'}{\mu} dt = \int \frac{2}{t} dt$$

$$\ln |\mu| = 2 \ln |t| + C$$

$$e^{\ln |\mu|} = e^{\ln |t|^2 + C} = e^{\ln t^2} \cdot e^C$$

$$\mu(t) = t^2 \cdot C, \quad C \neq 0$$

$$t^2 y' + \frac{2}{t} t^2 y = \frac{\cos t}{t^2} \cdot t^2$$

$$(\mu y)' = (t^2 y)' = \cos t$$

$$\int (t^2 y)' dt = \int \cos t dt$$

$$t^2 y = \sin t + C$$

$$y = \frac{\sin t}{t^2} + C t^{-2} \rightarrow \text{gen. soln.}$$

$$\text{IV: } y(\pi) = 0 \Rightarrow y(\pi) = \frac{\sin \pi}{\pi^2} + \frac{C}{\pi^2} = 0$$

$$\cancel{\sin \pi} + C = 0 \Rightarrow C = 0$$

$$y(t) = \frac{\sin t}{t^2}$$

30. Find the value of y_0 for which the solution of the initial value problem

$$y' - y = 1 + 3 \sin t, \quad y(0) = y_0$$

remains finite as $t \rightarrow \infty$.

$$\mu y' - \underbrace{\mu y}_{\mu' = -\mu} = (1 + 3 \sin t) \mu$$

$$\int \frac{\mu'}{\mu} = \int -1 \Rightarrow \ln |\mu| = -t + C$$
$$\mu = e^{-t}$$

$$(e^{-t} y)' = (1 + 3 \sin t) e^{-t}$$

$$\int \underbrace{(e^{-t} y)'}_{e^{-t} y} dt = \int \underbrace{e^{-t}}_{-e^{-t}} dt + 3 \int \underbrace{e^{-t} \sin t}_{\text{by parts}} dt \quad (\star)$$

$$\int \underbrace{e^{-t}}_u \underbrace{\sin t}_{dv} dt = -e^{-t} \cos t - \int \underbrace{\cos t}_{\frac{dv}{dt}} \underbrace{e^{-t}}_u dt$$

$$\boxed{\int u dv = uv - \int v du} = -e^{-t} \cos t - \left[e^{-t} \sin t + \int \sin t e^{-t} dt \right]$$

$$= -e^{-t} \cos t - e^{-t} \sin t - \int \sin t e^{-t} dt$$

$$\int e^{-t} \sin t dt = \frac{1}{2} (-e^{-t} \cos t - e^{-t} \sin t) + C$$

$$(\star) \quad e^{-t} y = -e^{-t} + \frac{3}{2} (-e^{-t} \cos t - e^{-t} \sin t) + C \quad (\cdot e^t)$$

$$y = -1 + \frac{3}{2}(-\cos t - \sin t) + C e^t \rightarrow \text{gen. soln.}$$

$$\underline{\text{IV:}} \quad y(0) = y_0 \Rightarrow y(0) = -1 + \frac{3}{2}(-1 - 0) + C e^0 = y_0$$

$$-\frac{5}{2} + C = y_0$$

$$C = y_0 + \frac{5}{2}$$

Soln to IVP:

$$y = -1 + \frac{3}{2}(-\cos t - \sin t) + \left(y_0 + \frac{5}{2}\right)e^t$$

Annotations for the expression above:

- For $-\cos t$: $-1 \leq \leq 1$
- For $-\sin t$: $-1 \leq \leq 1$
- For the sum $(-\cos t - \sin t)$: $-2 \leq \leq 2$
- For the term $\frac{3}{2}(-\cos t - \sin t)$: $-3 \leq \leq 3$
- For the entire expression $-1 + \frac{3}{2}(-\cos t - \sin t)$: $-4 \leq \leq 2$
- The term e^t is marked with $\rightarrow \infty$.

To kill the $(y_0 + \frac{5}{2})e^t$ part, choose $y_0 = -\frac{5}{2}$

so that as $t \rightarrow \infty$ y remains finite ($-4 \leq y(t) \leq 2$ everywhere)

In each of Problems 1 through 8, solve the given differential equation.

1. $y' = x^2/y$ 1st ord. nonlin.

6. $xy' = (1 - y^2)^{1/2}$

$$\frac{dy}{dx} dx$$

$$dy(x,t) = \frac{\partial y}{\partial x} dx + \frac{\partial y}{\partial t} dt$$

Careful!

here we lost the solutions
 $y=1$
 $y=-1$

① $\int y dy = \int x^2 dx$

$$\frac{y^2}{2} = \frac{x^3}{3} + C$$

$$y^2 = \frac{2x^3}{3} + C \rightarrow \text{implicit soln.}$$

$$y = \pm \sqrt{\frac{2x^3}{3} + C} \rightarrow \text{explicit soln.}$$

⑥ $x \frac{dy}{dx} = \sqrt{1-y^2}$

$$\int \frac{dy}{\sqrt{1-y^2}} = \int \frac{dx}{x}$$

$$\arcsin y = \ln|x| + C$$

$$y = \sin(\ln|x| + C)$$

23. Solve the initial value problem

$$y' = 2y^2 + xy^2, \quad y(0) = 1$$

and determine where the solution attains its minimum value.

$$\int \frac{dy}{y^2} = \int (2 + x) dx$$

$$-\frac{1}{y} = 2x + \frac{x^2}{2} + C$$

$$y = \frac{1}{-2x - \frac{x^2}{2} + C} = -\frac{2}{4x + x^2 + C} \rightarrow \text{gen. soln.}$$

IV: $y(0)=1 \Rightarrow y(0) = -\frac{2}{C} = 1 \Rightarrow C = -2$

Solu to IVP: $y(x) = -\frac{2}{4x+x^2-2} = -2(4x+x^2-2)^{-1}$

Min. of $y(x)$:

$$y'(x) = +2(4x+x^2-2)^{-2} \cdot (4+2x) = 0 \Rightarrow \begin{matrix} 4+2x=0 \\ x=-2 \end{matrix}$$

or = undef. $\Rightarrow x^2+4x-2=0$

$$(x+2)^2 = 6$$

$$x_{1,2} = -2 \pm \sqrt{6} \rightarrow \text{vertical asymptotes}$$

$$\lim_{x \rightarrow (-2+\sqrt{6})^+} y(x) = \lim_{x \rightarrow (-2+\sqrt{6})^+} -\frac{2}{\underbrace{x^2+4x-2}_{(x+2)^2-6}} = -\infty$$

$$x > -2+\sqrt{6}$$

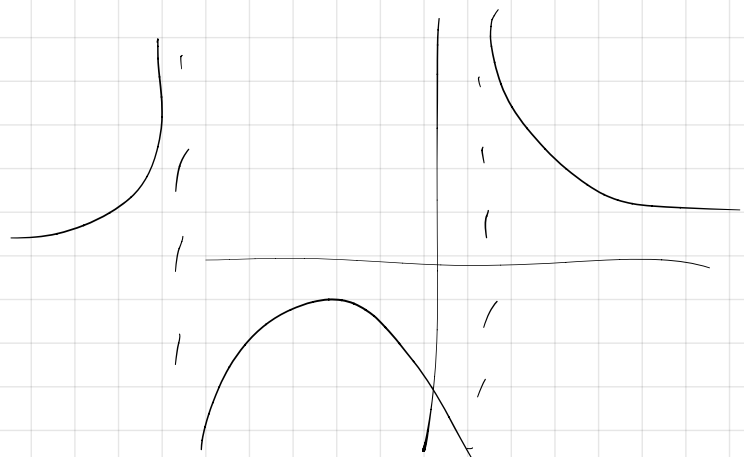
$$x+2 > \sqrt{6}$$

$$(x+2)^2 > 6$$

$$(x+2)^2 - 6 > 0$$

$y(x)$ doesn't have
a minimum.

Sketch the graph to check



30. Consider the equation

$$\frac{dy}{dx} = \frac{y-4x}{x-y} \stackrel{?}{=} \frac{2y-4x}{2x-2y} \quad (i)$$

(a) Show that Eq. (i) can be rewritten as

$$\frac{dy}{dx} = \frac{(y/x)-4}{1-(y/x)} = F\left(\frac{y}{x}\right) \quad (ii)$$

$$F(v) = \frac{v-4}{1-v}$$

thus Eq. (i) is homogeneous.

(b) Introduce a new dependent variable v so that $v = y/x$, or $y = xv(x)$. Express dy/dx in terms of x , v , and dv/dx .

(c) Replace y and dy/dx in Eq. (ii) by the expressions from part (b) that involve v and dv/dx . Show that the resulting differential equation is

$$v + x \frac{dv}{dx} = \frac{v-4}{1-v} \quad -v$$

or

$$x \frac{dv}{dx} = \frac{v^2-4}{1-v} \quad (iii)$$

Observe that Eq. (iii) is separable.

(d) Solve Eq. (iii), obtaining v implicitly in terms of x .

(e) Find the solution of Eq. (i) by replacing v by y/x in the solution in part (d).

homogeneous

$$y' = F\left(\frac{y}{x}\right) = F\left(\frac{\lambda y}{\lambda x}\right)$$

$$\boxed{v = \frac{y}{x}}$$

$$vx = y$$

$$v'x + v = y'$$

$$(d) \int \frac{1-v}{v^2-4} dv = \int \frac{dx}{x} = \ln|x| + c$$

$$\frac{1-v}{v^2-4} = \frac{1-v}{(v-2)(v+2)} = \frac{A}{v-2} + \frac{B}{v+2} = \frac{Av+2A+Bv-2B}{v^2-4}$$

$$\begin{cases} 1 = 2A - 2B \\ -1 = A + B \\ -2 = 2A + 2B \end{cases} \Rightarrow -1 = 4A \Rightarrow A = -\frac{1}{4}, B = -\frac{3}{4}$$

$$\ln|x| + c = \int \frac{1-v}{v^2-4} dv = \int -\frac{1}{4} \left(\frac{1}{v-2} \right) - \frac{3}{4} \left(\frac{1}{v+2} \right) dv$$

$$= -\frac{1}{4} \ln|v-2| - \frac{3}{4} \ln|v+2|$$

$$e^{\ln|x|+c} = e^{-\frac{1}{4} \ln|v-2| - \frac{3}{4} \ln|v+2|}$$

$$e^{\ln|x|} \cdot e^c = e^{\ln|v-2|^{-\frac{1}{4}} + \ln|v+2|^{-\frac{3}{4}}} = e^{\ln|v-2|^{-\frac{1}{4}}} \cdot e^{\ln|v+2|^{-\frac{3}{4}}}$$

$$|x| \cdot c_2 = |v-2|^{-\frac{1}{4}} \cdot |v+2|^{-\frac{3}{4}}$$

$$C^4 \cdot x^4 = |v-2|^{-1} \cdot |v+2|^{-3}$$

$$v = \frac{y}{x} \quad \text{back substitute}$$

$$x^4 = \frac{C}{\left| \frac{y}{x} - 2 \right| \cdot \left| \frac{y}{x} + 2 \right|^3}$$

$$= \frac{C}{\left| \frac{y-2x}{x} \right| \cdot \left| \frac{y+2x}{x} \right|^3}$$

$$x^4 = \frac{C x^4}{|y-2x| \cdot |y+2x|^3}$$

$$|y-2x| \cdot |y+2x|^3 = C$$

$$(y-2x)(y+2x)^3 = C$$

Ex & Un.
for
1st ord.
lin. ODE

Theorem 2.4.1 If the functions p and g are continuous on an open interval $I: \alpha < t < \beta$ containing the point $t = t_0$, then there exists a unique function $y = \phi(t)$ that satisfies the differential equation

$$1 \quad y' + p(t)y = g(t) \quad (1)$$

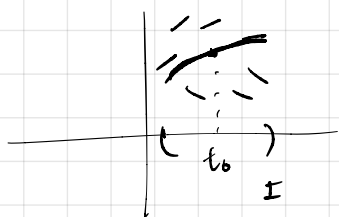
for each t in I , and that also satisfies the initial condition

$$y(t_0) = y_0. \quad (2)$$

where y_0 is an arbitrary prescribed initial value.

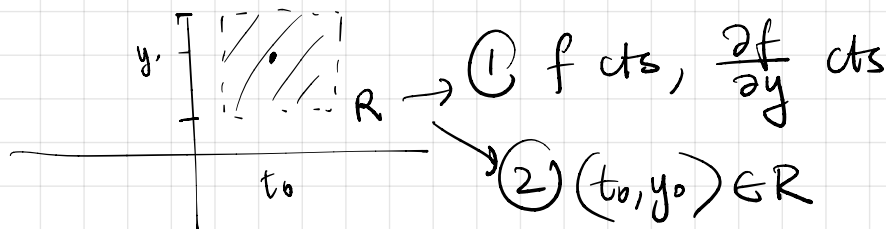
A soln y exists & unique on $I \rightarrow$ ① $t_0 \in I$

$\rightarrow$ ② $p(t), g(t)$ cts on I



Theorem 2.4.2 Let the functions f and $\partial f / \partial y$ be continuous in some rectangle $\alpha < t < \beta$, $\gamma < y < \delta$ containing the point (t_0, y_0) . Then, in some interval $t_0 - h < t < t_0 + h$ contained in $\alpha < t < \beta$, there is a unique solution $y = \phi(t)$ of the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0. \quad (9)$$



① f cts, $\frac{\partial f}{\partial y}$ cts

② $(t_0, y_0) \in R$

In each of Problems 1 through 6, determine (without solving the problem) an interval in which the solution of the given initial value problem is certain to exist.

4. $(4 - t^2)y' + 2ty = 3t^2, \quad y(-3) = 1$

① $-3 \in I$

② $1 \quad y' + \underbrace{\frac{2t}{4-t^2}}_{p(t)} y = \underbrace{\frac{3t^2}{4-t^2}}_{g(t)}$

$t = \pm 2$ discts

Continuity

f, g cts

• $f + g$ cts

• $f \cdot g$ cts

• f/g cts $\Leftrightarrow g \neq 0$

Then on $I = (-\infty, -2)$ a soln to the initial value prob. exists & it's unique.

Bernoulli Equations. Sometimes it is possible to solve a nonlinear equation by making a change of the dependent variable that converts it into a linear equation. The most important such equation has the form

$$y' + p(t)y = q(t)y^n,$$

and is called a Bernoulli equation after Jakob Bernoulli. Problems 27 through 31 deal with equations of this type.

27. (a) Solve Bernoulli's equation when $n = 0$; when $n = 1$.

(b) Show that if $n \neq 0, 1$, then the substitution $v = y^{1-n}$ reduces Bernoulli's equation to a linear equation. This method of solution was found by Leibniz in 1696.

$$\begin{aligned} (a) \quad y' + p(t)y &= q(t) \quad (n=0) \\ y' + p(t)y &= q(t)y \quad (n=1) \\ y' + (p(t)-q(t))y &= 0 \end{aligned}$$

$$(b) \quad n \geq 2: \quad y' + p(t)y = q(t)y^n \quad (\cdot y^{-n})$$

$$\underbrace{y' \cdot y^{-n}}_{v'/(1-n)} + p(t) \underbrace{y^{1-n}}_v = q(t)$$

$$v(t) = y^{1-n}(t)$$

$$v' = (1-n) y^{-n}(t) y'(t)$$

$$\frac{v'}{(1-n)} = y^{-n}(t) y'(t)$$

$$\left(v = y^{1-n} \right) \Rightarrow \frac{v'}{1-n} + p(t)v = q(t)$$

$$v' + p(t)(1-n)v = q(t)(1-n) \rightarrow 1^{st} \text{ ord. lin.}$$

use integrating factor

In each of Problems 28 through 31, the given equation is a Bernoulli equation. In each case solve it by using the substitution mentioned in Problem 27(b).

28. $t^2 y' + 2ty - y^3 = 0, \quad t > 0$

$$t^2 y' + 2ty = y^3$$

$$y' + \underbrace{\frac{2}{t}}_{p(t)} y = \underbrace{t^{-2}}_{q(t)} y^3 \xrightarrow{n=3} \Rightarrow \text{Bernoulli}$$

$$v = y^{-2}$$

$$v' = -2y^{-3} \cdot y'$$

$$\underbrace{y' \cdot y^{-3}}_{\substack{v' \\ -2}} + \frac{2}{t} \underbrace{y^{-2}}_v = t^{-2}$$

$$\Rightarrow \frac{v'}{-2} + \frac{2}{t} v = t^{-2} \quad (\cdot (-2))$$

$$\mu v' - \frac{4}{t} \mu v = -2t^{-2} \mu \rightarrow 1^{\text{st}} \text{ ord, lin.}$$

$$\mu' = -\frac{4}{t} \mu$$

$$\frac{\mu'}{\mu} = -\frac{4}{t}$$

$$C + \ln |\mu| = -4 \ln |t| = \ln |t|^{-4} \xrightarrow{\text{exp.}} \mu = C t^{-4}$$

$$\underbrace{t^{-4} v' - \frac{4}{t} t^{-4} v}_{(\mu v)'} = -2 t^{-2} \cdot t^{-4}$$

$$\int (t^{-4} \cdot v)' dt = \int -2 t^{-6} dt$$

$$t^{-4} \cdot v = \frac{2}{5} t^{-5} + C \quad (\cdot t^4)$$

$$v = \frac{2}{5} t^{-1} + C \cdot t^4$$

$$v = y^{-2}$$

$$\frac{1}{y^2} = y^{-2} = \frac{2}{5} t^{-1} + C t^4 = \frac{2}{5t} + C t^4 = \frac{2 + C t^5}{5t}$$

$$\frac{5t}{2 + C t^5} = y^2 \rightarrow \text{implicit soln.}$$

$$\pm \sqrt{\frac{5t}{2 + C t^5}} = y \rightarrow \text{expl.}$$